HW-6 : Dimensionality Reduction

MA 5755 - Data Visualization Ma25M004

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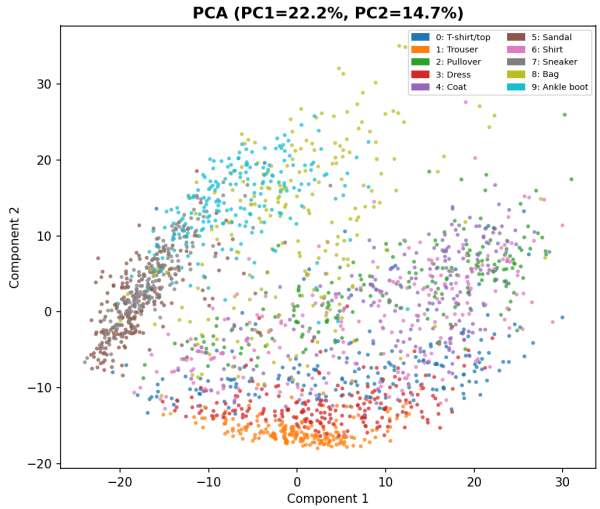
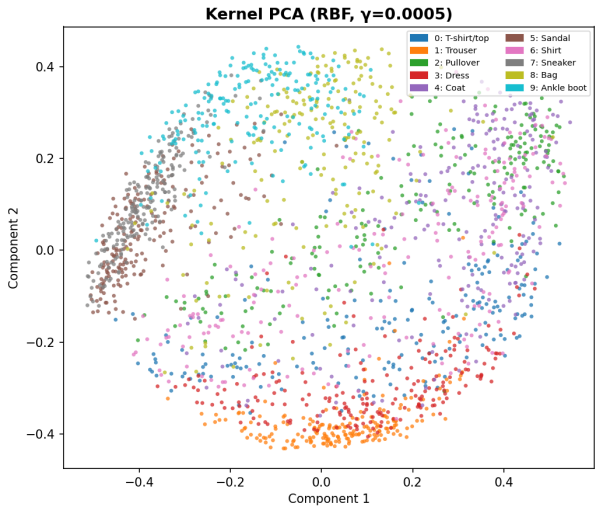
Objective

In this report we compare four dimensionality reduction methods PCA, Kernel PCA (RBF), t-SNE, and UMAP applied to a balanced 2 000-sample subset (200 per class) of the **Fashion-MNIST** dataset. Each sample is a 28×28 grayscale image of a clothing item, represented as a vector in R^{784}. All methods embed the data into R^2 for visual inspection. The emphasis is on the geometric structure of the resulting embeddings, not on the implementation details.

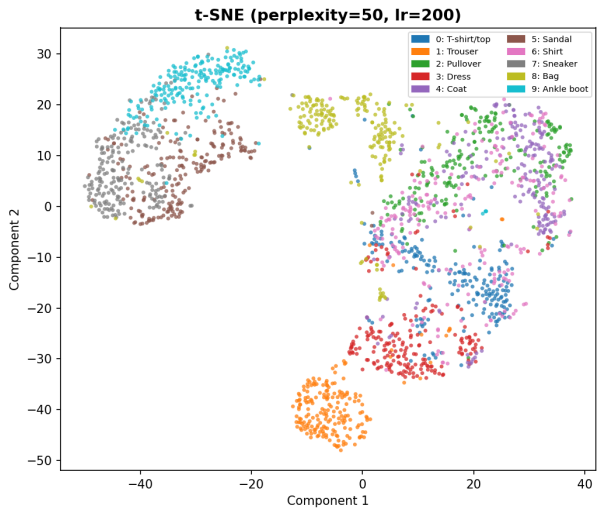
1 Embeddings and Observations



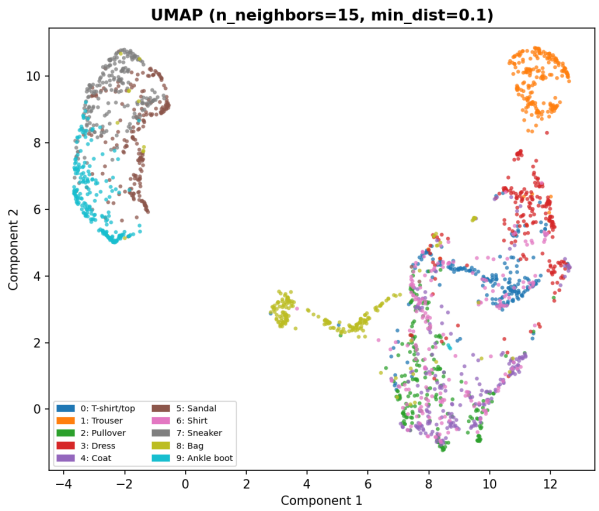
(a) PCA linear projection onto the top 2 principal components.



(b) Kernel PCA RBF kernel with $\gamma = 0.0005$.



(c) t-SNE perplexity= 50, learning rate= 200.



(d) UMAP $n_neighbors = 15$, $min_dist = 0.1$.

Figure 1: Two-dimensional embeddings of 2 000 Fashion-MNIST samples. Each colour corresponds to one of the 10 clothing categories (see legend).

1.1 PCA (Figure 1a)

PCA keeps the directions with the most overall variation, but two components keep only about 20–25percent information. Trouser and Bag form clear groups, while other clothes overlap. PCA shows global differences well but cannot capture complex nonlinear patterns in images.

1.2 Kernel PCA (Figure 1b)

Kernel PCA with RBF slightly improves clusters. Trouser and Bag separate well, but many clothes still overlap. It captures some nonlinearity but costs more. **Key insight:** Kernel PCA introduces nonlinearity but at significant computational cost ($O(n^2)$ in memory), and the improvement in cluster separation over linear PCA is modest for this dataset.

1.3 t-SNE (Figure 1c)

t-SNE keeps nearby points close when reducing dimensions. With perplexity 50 and learning rate 200, all classes form clear clusters. Similar clothes separate better, but distances between clusters do not represent real meaning. **Key insight:** t-SNE is best at revealing *local cluster structure* but sacrifices global geometry and is the slowest method.

1.4 UMAP (Figure 1d)

UMAP keeps the structure of high-dimensional data when reducing dimensions. With neighbors 15 and $\min_d ist 0.1, clustersareclear. Footwearappearsononeside, topsonanother. Itisfasterthant-SNEandgoodforexploringdata.$

Key insight:$UMAPsimultaneouslyachievescompactlocalclustersandagloballymeaningfullayout.$

2 Runtime Comparison

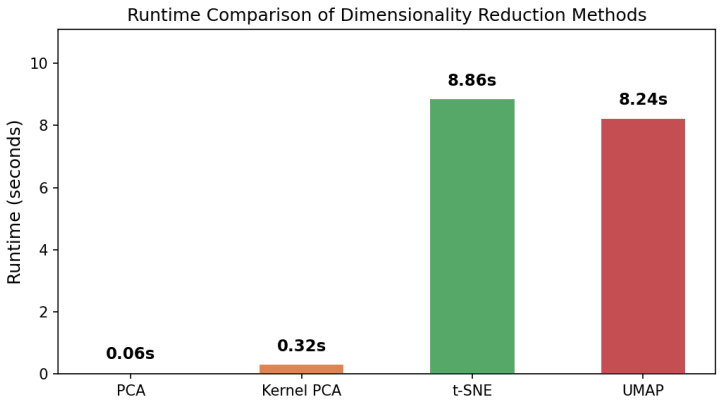


Figure 2: Wall-clock runtime for each method on the 2 000-sample subset.

PCA is by far the fastest method t-SNE is the slowest ,UMAP offers an excellent speed quality trade-off.

Criterion	PCA	Kernel PCA	t-SNE	UMAP
Cluster separation	Weak	Moderate	Excellent	Excellent
Global structure	Best	Good	Poor	Good
Local structure	Poor	Moderate	Excellent	Excellent
Speed	Fastest	Slow	Slowest	Fast
Inter-cluster distances interpretable	Yes	Partial	No	Partial
Scales to large n	Yes	No	Moderate	Yes

Table 1: Summary comparison of the four dimensionality reduction methods.

3 Conclusion and Method Selection

Recommended method: UMAP.

For Fashion-MNIST, UMAP is the superior single choice. It produces cluster separation comparable to t-SNE (all 10 clothing categories are visibly distinct) while additionally encoding meaningful global topology related garment types appear spatially adjacent in the embedding. This richer geometric structure makes the visualisation more informative and actionable: one can immediately see, for instance, that the footwear cluster sits apart from the tops cluster, with dresses and trousers forming a bridge. Furthermore, UMAP is roughly 4× faster than t-SNE and scales more gracefully to larger datasets, making it the practical choice for both exploratory analysis and downstream tasks such as cluster-based labelling or semi-supervised learning.

PCA remains indispensable as a fast pre-processing step (the PCA-50 reduction used before t-SNE and UMAP both reduces noise and speeds up those methods). Kernel PCA offers little additional benefit here at significant computational cost, and t-SNE, while producing excellent local clusters, discards global relationships that are scientifically meaningful for a dataset with a clear categorical hierarchy like clothing.